

mtcars (mtcars)

	mpg	wt	disp	hp	drat	wt	qsec	vs	am	gear	carb
Mazda RX4	21.0	2.62	264	150	3.21	2.875	16.99	0	1	4	4
Mazda RX4 Wag	21.0	2.875	264	150	3.21	2.875	16.99	0	1	4	4
Datsun 710	22.8	2.14	168	110	3.08	3.215	19.44	1	0	3	2
Hornet 4 Drive	21.4	2.6	253	110	3.08	3.215	19.44	1	0	3	2
Hornet Sportabout	18.7	2.77	360	175	3.15	3.44	19.44	0	0	3	1
Volvo 740 G	18.1	2.25	105	105	2.76	3.46	19.44	1	0	3	1

Ex: No: 7	Implementation of Linear Regression Using mtcars Dataset
Date: 29/4/26	

Problem Description:

The mtcars dataset contains specifications of different car models, including fuel efficiency and engine characteristics. To understand how engine and vehicle attributes influence fuel efficiency, linear regression can be used to model the relationship between **miles per gallon (mpg)** and predictors such as **engine displacement (disp)**, **horsepower (hp)**, and **weight (wt)**.

Objective:

To implement a linear regression model in R to analyze the relationship between fuel efficiency and vehicle attributes, interpret the regression coefficients and model summary, and visualize the regression results for better understanding of the model.

Procedure:

- Step 1: Load Required Libraries

We load the necessary libraries for implementing linear regression.

```
# Load required libraries
```

```
library(lme4)
```

```
library(ggplot2)
```

- Step 2: Load and Prepare the Dataset

We will use the built-in mtcars dataset and explore it.

```
# Load the dataset
```

```
data(mtcars)
```

```
# Display first few rows
```

```
head(mtcars)
```

Dataset Explanation:

mpg (Miles per gallon) → Dependent Variable (Y)

hp (Horsepower) → Independent Variable (X)

We will predict mpg based on hp using linear regression

Summary (model - simple)

Call:

lm (formula = mpg ~ hp, data = train_data)

Particulars:

min	1st	median	3rd	max
-5.0566	-2.0595	-0.8412	1.2261	8.8266

Coefficients:

Intercept	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	30.09557	1.82293	16.464	4.432e-06
hp	-0.08141	0.01191	-5.996	4.92e-06

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.9 on 22 degrees of freedom

Multiple R-squared: 0.629 Adjusted R-squared: 0.6031

F-statistic: 35.95 on 1 and 22 DF, p-val: 4.92e-06

Model (Predicted value)

Model	Actual	Predicted
Mazda RX4	21.0	23.45449
Mazda RX4 Sport	26.0	22.24053
Datsun 710	15.2	23.45449
Toyota Corolla	18.7	22.24053
Volvo 740 G	14.4	23.45449
Volvo 760	15.2	22.24053

- Step 3: Split the Dataset into Training and Testing Sets

We split the dataset into 70% training and 30% testing.

Split data into training (70%) and testing (30%)

```
train_index <- createDataPartition(mtcars$mpg, p=0.7,
                                     times=FALSE)
```

```
train_data <- mtcars[train_index, ]
test_data <- mtcars[-train_index, ]
```

- Step 4: Simple Linear Regression (mpg vs hp)

We build the linear regression model using lm() function.

Train linear regression model

```
lm_model <- lm(mpg ~ hp, data=train_data)
```

Print model summary

```
summary(lm_model)
```

- Step 5: Make Predictions on Test Data

We use the trained model to make predictions on the test dataset.

Make predictions on test data

```
predictions <- predict(lm_model, test_data)
```

Display predicted values

```
pred (predictions)
```

Linear Regression: HP vs MPG



Summary (model - multiple)

call:

$\ln$ (formula = mpg ~ wt + hp + disp, data = train_data)

Residuals:

min	1Q	median	3Q	max
-4.1639	-1.6123	-0.0093	1.3078	5.9539

• Step 6: Evaluate Model Performance

We compute Mean Squared Error (MSE) and R-squared for evaluation.

Compute Mean Squared Error (MSE)

$mse \leftarrow mean((text_data\$mpg - predictions)^2)$

$cat("Mean Squared Error:", mse, "\n")$

Compute R-squared value

$r_squared \leftarrow cor(text_data\$mpg, predictions)^2$

$cat("R-squared:", r_squared, "\n")$

• Step 7: Visualize Classification Results

We plot the regression line on a scatter plot.

Scatter plot with regression line - hp vs mpg

$ggplot(train_data, aes(x = hp, y = mpg)) +$

$geom_point(aes(color = "blue", size = 3)) +$

$geom_smooth(method = "lm", aes(linetype = "dashed")) +$

$ggtitle("Linear Regression: Horsepower vs MPG")$

• Step 8: Multiple Linear Regression - mpg ~ wt, hp, disp

$model_multiple \leftarrow \ln(mpg \sim wt + hp + disp,$

$data = train_data)$

Model summary

Summary(model_multiple)

Coefficients:

	Estimate	Std. Error	t-value	p(> t)
(Intercept)	38.974631	2.593486	15.028	2.33e-12 ***
wt	-4.741613	1.342959	-3.531	0.00011 ***
hp	-0.030924	0.012858	-2.358	0.0290 *
drat	-0.063784	0.011662	0.324	0.7490

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.616 on 80 degrees of freedom

Multiple R-Squared: 0.8447, Adjusted R-Squared: 0.8214.

F-statistic: 36.27 on 3 and 200 df, P-value: 2.794e-08

→ head (predicted - multiple)

Mazda RX4	Datsun 7TD	Hornet 4	Drive Hornet	Sebring
83.83241	26.57193	81.38198	18.73652	

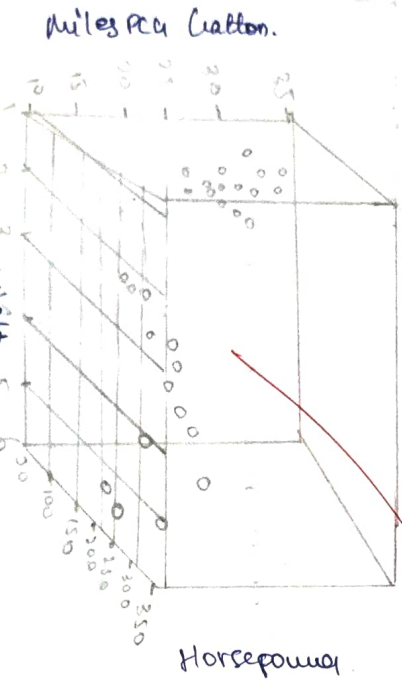
Volant

more 2400

00.24653

22-53011

3D Scatter plot: mpg vs weight and hp.



- Step 9: Multiple Linear Regression - Prediction

Predicted mpg values

predicteel - multiple ← predicteel (model - multiple)

head (predicteel - multiple)

- Step 10: 3D Visualization (mpg vs wt and hp)

library (scatterplot3d)

scatterplot3d (mtcars, x = wt, y = hp, z = mpg)

xlab = "Weight"

ylab = "Horsepower"

zlab = "MPG"

main = "3D Plot of MPG vs Weight and Horsepower"

Result:

Thus, the regression analysis demonstrates that vehicle weight, horsepower, and engine displacement have a significant negative effect on fuel efficiency. The multiple linear regression model provides better explanatory power than simple regression and successfully predicts miles per gallon based on vehicle characteristics.